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/**
 * server/lib/safe-json-parser.ts
 *
 * Fix #5 – content-grinder JSON Truncated at Position 15,494
 *
 * The AI returns JSON that gets cut off mid-string when max_tokens is hit.
 * Standard JSON.parse throws on truncated input. This utility attempts
 * recovery before giving up – important for the content-grinder's moment
 * extraction which processes large video catalogs.
 */

import { createLogger } from './logger.js';

const log = createLogger('safe-json-parser');

/**
 * Attempts to parse potentially truncated JSON from an AI response.
 * Tries multiple recovery strategies before returning null.
 */
export function safeParseAiJson<T = unknown>(
  raw: string,
  label?: string,
): T | null {
  if (!raw || raw.trim().length === 0) return null;

  // Clean common AI response wrappers
  const cleaned = raw
    .replace(/^```json\s*/i, '')
    .replace(/^```\s*/i, '')
    .replace(/\s*```$/, '')
    .trim();

  // Attempt 1 – clean parse
  try {
    return JSON.parse(cleaned) as T;
  } catch { /* fall through */ }

  // Attempt 2 – truncated object: close with }
  try {
    const recovered = cleaned.replace(/,\s*$/, '') + '}';
    return JSON.parse(recovered) as T;
  } catch { /* fall through */ }

  // Attempt 3 – truncated array: close with ]
  try {
    const recovered = cleaned.replace(/,\s*$/, '') + ']';
    return JSON.parse(recovered) as T;
  } catch { /* fall through */ }
}

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// Attempt 4 – truncated nested: close both
try {
  const recovered = cleaned.replace(/,\?\s*$/, '') + '{}';
  return JSON.parse(recovered) as T;
} catch { /* fall through */ }

// Attempt 5 – find last complete top-level key and truncate there
try {
  // Find the last complete "key": value pair
  const lastComplete = cleaned.lastIndexOf('"',');
  if (lastComplete > 10) {
    const truncated = cleaned.slice(0, lastComplete + 1) + '{}';
    return JSON.parse(truncated) as T;
  }
} catch { /* fall through */ }

log.warn(
  `[SafeJsonParser]${label ? `[${label}]` : ''} All recovery attempts failed. ` +
  `Raw length: ${raw.length}. First 100 chars: ${raw.slice(0, 100)}`
);
return null;
}

/**
 * Recommended max_tokens values for calls that return JSON.
 * The content-grinder moment extraction was hitting 15,494 chars
 * (~3,870 tokens) with a likely max_tokens of 4096. Increase to 6000.
 */
export const AI_JSON_TOKEN_LIMITS = {
  momentExtraction: 6000, // was hitting 4096 limit – this is why JSON was truncated
  titleGeneration: 200,
  descriptionGen: 600,
  tagGeneration: 250,
  thumbnailConcept: 400,
  chapterGeneration: 300,
  strategyScoring: 150,
  clipScoring: 300,
} as const;

```